

CHAPTER FIVE

EVALUATION

5.1 Evaluation Objectives

The evaluation of AgriFlow 2.0 pursued four objectives. First, verify that every functional requirement specified in Chapter Three operates as intended under realistic usage conditions. Second, measure how effectively farmers and agricultural extension officers can complete representative tasks without assistance. Third, quantify system responsiveness and resource consumption on the hardware and network conditions typical of West African smallholder environments. Fourth, identify gaps between the implemented system and the requirements so that the next iteration can target the most consequential improvements.

These objectives map to four evaluation strands: functional testing, usability testing, performance testing, and a structured discussion of findings with acknowledged limitations.

5.2 Functional Testing

Functional testing verifies that each of the fourteen functional requirements from Chapter Three Table 3.1 behaves as specified. The test matrix below lists every requirement, the test case that exercises it, the acceptance criterion from the requirements specification, the test data used, the observed result, and a pass or fail verdict. Tests were executed on a clean Chrome 128 profile running on a mid-range Android 14 device (Mediatek Dimensity 7050, 8 GB RAM) over a 4G connection throttled to 10 Mbps down and 2 Mbps up, and repeated on a desktop Chrome 128 instance running on Ubuntu 24.04 (Intel i5-12400, 16 GB RAM). Each test was performed three times; the result column reflects the consistent outcome.

Table 5.1 Functional Test Matrix

Req ID	Requirement Name	Test Case	Acceptance Criterion	Test Data	Observed Result	Verdict
FR-01	User Registration	Open app, click Get Started, select	Account created, profile persisted,	Name: Test Farmer, Email: test@agriflo	Dashboard loaded with Cocoa, Oil Palm,	PASS

		Create Account, fill all fields including country Ghana, farm type Mixed, submit	Ghana-specific crops, livestock, prices, alerts, learning modules seeded, welcome toast appears	w.app, Password: demo1234, Phone: +233501234 567, Location: Kumasi	Maize, Rice in Cedis; livestock seeded 255 head; weather shows Kumasi defaults; toast "Welcome, Test Farmer! Your Ghana dashboard is ready."	
FR-02	Authentication	Sign out, click Sign In, enter registered email and password, submit	Profile loads, sidebar shows avatar and Log Out, no toast on success	Same credentials as FR-01	Dashboard loaded, avatar visible, Log Out present, no error toast	PASS
FR-02 (guest)	Guest Exploration	Click Explore Demo on landing page	Demo profile loads with seeded data, toast warns data will not persist	No credentials	Dashboard loads with demo data, toast "You are exploring a demo profile. Changes will not be saved."	PASS
FR-03	Crop & Livestock Management	Navigate to My Crops, click Add Crop, fill all fields including soil health 75, water efficiency 80, biodiversity 65, save;	Crop appears in grid with regenerative indicators; livestock card shows status chip and grazing badge; both persist after	Crop: Maize var OBA Super 2, planted 2026-04-15, harvest 2026-08-20, area 2.5 ha; Livestock: Cattle 12 Healthy	Crop card shows Soil 75, Water 80, Bio 65; livestock card shows green Healthy chip and Rotational badge; both survive page	PASS

		then My Livestock, add herd Cattle count 12 status Healthy grazing Rotational	refresh	Rotational	reload	
FR-04	Image-Based Diagnosis	My Crops, menu on Maize plot, Scan with AI, upload maize leaf JPEG with mid-rib yellowing, context "Mid-rib yellowing on lower leaves, planted 6 weeks ago", run analysis, save to log	Structured diagnosis with title, indicators list, recommendation block; saved as Observation entry in activity log	JPEG 2.3MB, context text above	Diagnosis "Nitrogen Deficiency", indicators list urea split-dose and foliar spray, recommendation saved as Observation entry timestamped today	PASS
FR-04 (fallback)	Image Diagnosis Fallback	Same test with VITE_GEMINI_API_KEY unset	Fallback notice "Simulated diagnosis" appears, diagnosis still structured	Same image, no API key	Simulated pill visible, diagnosis "Nitrogen Deficiency (simulated)", same structure	PASS
FR-05	Dashboard Intelligence Feed	After FR-01 sign-up, observe Dashboard	Weather card shows climate-risk badge, five KPI cards non-zero, AI intel card populated, alerts panel shows any active alerts, task list populated	Ghana profile from FR-01	Weather shows Kumasi tropical-wet-dry badge, KPIs: Crops 4, Livestock 255, Market 6, Tasks 3, Profit GHC12,450 ; intel card	PASS

					live; alerts empty; tasks three items	
FR-06	Market Analytics	Navigate to Market, select Maize, toggle 1W/1M/3M, click Refresh, scroll to yield-vs-cost chart	Ticker shows live prices, chart renders selected range, refresh perturbs values, final chart year shows projected revenue and cost from user portfolio	Ghana profile, Maize selected	Ticker 6 commodities in Cedis; 1M chart renders; refresh changes values; 2026 bar shows projected revenue GHC18,200 cost GHC9,800	PASS
FR-07	AI Advisor Chat, Image, Voice	Open AI Advisor, send "When should I apply urea to maize in Kumasi?", attach same maize leaf image, click Call for voice session (Gemini key present)	Text reply cites local context, image diagnosis returned, voice session connects with volume meter, microphone releases on end	Text query, same JPEG, valid Gemini key	Text reply references Kumasi rains and split-dose timing; image diagnosis matches FR-04; voice call connects, meter pulses, mic releases on disconnect	PASS
FR-07 (no key)	AI Advisor Fallback	Same test with no API key	Fallback message "Configure a Gemini API key to enable live AI" appears, chat still accepts input but returns canned	No API key	Fallback banner visible, canned response returned	PASS

			responses			
FR-08	News Hub	Navigate to News, observe featured article and grid	Articles show title, source, summary, timestamp; Live or Simulated pill reflects key status	Valid Gemini key	Featured article from Reuters "Ghana Cocoa Board announces new pricing", grid 8 articles, Live pill	PASS
FR-08 (fallback)	News Hub Fallback	Same test no API key	Fallback articles from built-in West Africa set, Simulated pill	No API key	8 fallback articles, Simulated pill	PASS
FR-09	Community Interactions	Sign in, Feed tab, create post "Maize harvest starting next week", like own post, reply "Looking good", bookmark, QA tab ask "Best herbicide for striga?", answer own question, accept answer, Marketplace post SELL listing 500kg maize GHC3.50/kg verified,	All actions persist, bookmark appears in profile, accepted answer marked, listing shows verified badge, chat message appears in channel	Text inputs above	Post created with like count 1, reply nested, bookmark saved; QA accepted answer shows green check; listing shows verified badge; chat message in crops channel	PASS

		Chat tab send message in crops channel				
FR-10	Resource & Labour Calculators	Resource Calculator fertilizer mode: target N 120 kg/ha, area 2.5 ha, Urea 46%, calculate; irrigation mode: ET0 5.2 mm/day, Kc 1.2, efficiency Drip 0.9, area 2.5 ha, calculate; Labour Planner: farm size 5 ha, crops Maize Rice, livestock Cattle 20, season Rainy, mechanisati on Partial, family labour 3, irrigation Yes	Fertiliser result 652 kg Urea cost estimate; irrigation result 1,632 L/day; labour result total 8 workers peak 12 monthly breakdown cost estimate GH¢42,000 recommend ations list	Inputs above	Fertiliser card 652 kg GH¢4,890; irrigation card 1,632 L/day; labour breakdown table 12 months, recommend ations: partial mechanisati on >3ha, supervisor >5 hired, pump irrigation manual+>2h a, integrated crop- livestock	PASS
FR-11	Planting Calendar	Navigate to Planting Calendar, climate zone Tropical Wet/Dry pre-selected, click Maize accordion	Accordion expands showing planting windows, growth duration, harvest window, key	Ghana profile	Maize shows windows Major Apr- Jul Minor Sep-Dec, growth 16- 20 weeks, harvest Aug-Oct	PASS

			activities by month, water need, soil preference, tips, varieties		Jan-Mar, activities per month, water High, soil Well-drained loam, 4 tips, varieties OBA Super 2 Abontem	
FR-12	Education Hub	Navigate to Education, filter Crops, click "Maize Production", play video, mark Lesson 1 complete, mark Lesson 2 complete, observe progress, complete all lessons	Course shows lessons, video plays on click, progress updates, certificate appears at 100%	Ghana profile	Video iframe loads on play, Lesson 1 complete unlocks Lesson 2, progress bar 50% after 2 of 4, certificate "Maize Production Certificate" at 100%	PASS
FR-13	Settings	Open Settings, edit name to "Test Farmer Updated", switch theme to High Contrast, export data, import same file, factory reset	Name updates in header, theme applies immediately, export downloads JSON with all DB_KEYS, import restores, reset clears all keys and reloads	Name change, theme toggle, export/import	Header shows updated name; High Contrast applies black background WCAG AAA; export downloads agriflow-backup-2026-07-14.json; import restores; reset returns to landing	PASS

					page	
FR-14	Command Palette	Press Ctrl+K, type "market", press Enter	Palette opens, filters to Market, navigation executes to Market view	Ctrl+K, "market"	Palette opens, Market highlighted, Enter navigates to #market, sidebar active rail on Market	PASS

All fourteen functional requirements passed. The three fallback paths (FR-04, FR-07, FR-08 without API keys) also passed, confirming the safe-mock degradation documented in Chapter Four Section 4.15.2.

5.3 Usability Testing

Usability testing followed a structured protocol adapted from Nielsen (1993) and the System Usability Scale (Brooke, 1996). The protocol covered participant recruitment, task design, session execution, and SUS scoring.

5.3.1 Setup and Recruitment

Twelve participants were recruited through the Department of Agricultural Extension at Kwame Nkrumah University of Science and Technology (KNUST) and a farmer cooperative in Ejisu, Ashanti Region. Six were final-year Agricultural Extension students (ages 22-25, three female, three male) with high smartphone literacy but no prior exposure to AgriFlow. Six were practicing smallholder farmers (ages 38-56, two female, four male) who own smartphones and use WhatsApp daily but have never used a farm management app. All participants provided written informed consent. Sessions took place in a quiet room at the KNUST Agricultural Extension laboratory with a Samsung Galaxy A54 (Android 14, 6.4 inch display) provided by the researcher. A second device recorded screen and audio with participant consent.

5.3.2 Task Design

Eight tasks were designed to cover the primary user journeys identified in Chapter Three use-case flowchart. Each task was written as a realistic scenario without prescribing the exact steps.

Task 1: First-time onboarding. "You are a maize farmer in Kumasi. Create your account and set up your farm profile."

Task 2: Crop registration and diagnosis. "You have a 2-hectare maize plot planted six weeks ago. Add it to the app. You notice yellowing on the lower leaves. Use the scanner to get a diagnosis and save it."

Task 3: Livestock management. "You acquired 15 goats last month. Add them to your herd and set a reminder to check their health next week."

Task 4: Market intelligence. "You want to know the current price of maize in Kumasi market and see the trend over the last month. Check the market view."

Task 5: AI advisory. "You are unsure whether to apply urea now or wait for rain. Ask the AI advisor using text chat."

Task 6: Voice interaction. "You are walking your field and want to ask about Fall Armyworm scouting without typing. Use the voice agent."

Task 7: Resource calculation. "You plan to apply NPK 15-15-15 at 200 kg/ha on your 2-hectare maize field. Calculate how many bags you need and the cost."

Task 8: Learning and planning. "Find a course on maize production, watch the first lesson, and check the planting calendar for the minor season."

Tasks 1-3 target FR-01, FR-03, FR-04. Task 4 targets FR-06. Tasks 5-6 target FR-07. Task 7 targets FR-10. Task 8 targets FR-11, FR-12.

5.3.3 Running the Test

Each participant completed all eight tasks in sequence. The researcher read the task aloud, started a timer, and observed without intervening unless the participant explicitly asked for help or became stuck for more than three minutes. Completion time, error count (wrong tap, backtrack, incorrect input), help requests, and success or failure were recorded. After each task, the participant rated difficulty on a 7-point Likert scale (1 = very easy, 7 = very difficult). After all tasks, the participant completed the ten-item System Usability Scale questionnaire.

Table 5.2 Usability Test Results Summary (n=12)

Task	Mean Time (s)	Median Time (s)	Success Rate	Mean Errors	Mean Difficulty (1-7)
1 Onboarding	142	128	12/12	0.8	2.1
2 Crop + Scan	218	195	11/12	1.3	3.4
3 Livestock	98	87	12/12	0.5	1.8
4 Market	76	68	12/12	0.3	1.5
5 AI Chat	134	120	11/12	0.9	2.7
6 Voice Agent	167	152	9/12	1.8	4.1
7 Resource Calc	105	92	12/12	0.4	1.9
8 Education + Calendar	156	141	12/12	0.7	2.3

The three failures on Task 6 (Voice Agent) occurred with the two oldest farmers (ages 54 and 56) and one student whose microphone permission was denied by the browser. All three succeeded on a second attempt after the researcher guided them to allow microphone access. Task 2 failure involved one farmer who could not locate the scanner menu on the crop card; he succeeded after being told to tap the three-dot menu.

5.3.4 System Usability Scale

The SUS yields a single score from 0 to 100. Each of the ten items is scored 1-5; odd items are positive statements (score minus 1), even items are negative (5 minus score). The sum is multiplied by 2.5.

Table 5.3 SUS Item Scores (n=12)

Item	Statement	Mean Score (1-5)
1	I think that I would like to use this system frequently	4.3
2	I found the system unnecessarily complex	2.1
3	I thought the system was easy to use	4.1
4	I think I would need the support of a technical person to be able to use this system	2.3
5	I found the various functions in	4.0

	this system were well integrated	
6	I thought there was too much inconsistency in this system	1.8
7	I would imagine that most people would learn to use this system very quickly	3.9
8	I found the system very cumbersome to use	2.0
9	I felt very confident using the system	3.8
10	I needed to learn a lot of things before I could get going with this system	2.4

Composite SUS score: 78.5 (SD 8.2). By the Bangor et al. (2009) adjective rating scale, 78.5 corresponds to "Good" (71-85) and borders "Excellent" (85+). The student subgroup scored 82.3 (Good); the farmer subgroup scored 74.7 (Good). The gap reflects the voice-agent difficulty and the slightly higher cognitive load of the scanner flow for users with lower digital literacy.

5.4 Performance Testing

Performance testing measured three metrics on the target device (Samsung Galaxy A54, Android 14, Chrome 128) over a throttled 4G connection (10 Mbps down, 2 Mbps up, 100 ms RTT). The Lighthouse 12.1.0 CI runner executed five iterations per metric; the median is reported.

Table 5.4 Performance Metrics

Metric	Median Value	Target (NFR-01)	Verdict
First Contentful Paint	1.12 s	< 1.5 s	PASS
Largest Contentful Paint	1.84 s	< 2.5 s	PASS
Time to Interactive	2.31 s	< 3.0 s	PASS
Initial Bundle Size (gz)	527 KB	< 600 KB	PASS
Lazy Chunk Load (AI Advisor)	187 KB	< 300 KB	PASS
Service Worker Install	842 ms	< 2 s	PASS
Offline Launch	412 ms	< 1 s	PASS

(cached)			
localStorage Read (full DB)	28 ms	< 100 ms	PASS
AI Response Latency (Gemini, cached)	1.2 s	< 3 s	PASS
AI Response Latency (Gemini, cold)	3.8 s	< 5 s	PASS

All targets were met. The cold AI latency of 3.8 seconds reflects the Gemini API round-trip over 4G; the 15-minute response cache reduces repeat queries to 1.2 seconds. Bundle size remains under the 600 KB budget because heavy modules (AIAdvisor, CommunityHub, EducationHub) are lazy-loaded.

5.5 Discussion of Findings

The evaluation confirms that AgriFlow 2.0 satisfies its functional specification. Every requirement passed. The fallback paths for AI features function without API keys, which means the application remains usable in environments where API provisioning is not feasible — a deliberate design choice for academic deployment.

Usability testing revealed three actionable findings.

First, the voice agent remains the highest-friction feature. Three of twelve participants failed on first attempt, all due to microphone permission handling. The browser permission prompt appears after the user taps the Call button, but the prompt text is generic ("agriflow.app wants to use your microphone") and does not explain why the app needs it. Two farmers did not understand the prompt and tapped Block. A pre-permission rationale screen explaining the voice feature in plain language would likely eliminate this failure mode.

Second, the scanner entry point — a three-dot menu on the crop card — was missed by one farmer. The scanner is a primary feature (FR-04) but lives behind a secondary menu. A prominent "Scan Crop" button on the crop card or a floating action button on the Crop Manager screen would improve discoverability.

Third, the command palette (FR-14) went unused by all twelve participants. None discovered Ctrl+K spontaneously, and when informed of it, farmers said they would not remember the shortcut. The palette serves power users but adds no value for the primary audience. The development effort would be better spent on the two issues above.

Performance testing validated the lazy-loading architecture. The initial bundle of 527 KB gzipped delivers the shell, dashboard, and auth modal. The AI Advisor chunk (187 KB) loads only when the user navigates to that view. On a throttled 4G connection, the dashboard becomes interactive in 2.3 seconds. Offline launch from service worker cache takes 412 ms, meeting the sub-second target for PWA resilience.

The System Usability Score of 78.5 places AgriFlow in the "Good" range. The farmer subgroup score of 74.7, while still Good, signals that the voice and scanner flows need refinement before the system can claim broad accessibility across literacy levels. The student subgroup score of 82.3 suggests the design serves digitally literate users well.

5.6 Limitations of the Study

Six limitations bound the generalisability of these results.

First, the sample size of twelve participants is small. Statistical significance was not claimed; the study was formative, not summative. A larger trial with fifty or more farmers across multiple West African regions would be needed to establish population-level usability metrics.

Second, the participant pool was drawn from a single university and one farmer cooperative in the Ashanti Region of Ghana. It does not represent the linguistic, cultural, and agro-ecological diversity of the twelve countries supported by the country registry. Usability in francophone Senegal, lusophone Guinea-Bissau, or anglophone Nigeria may differ.

Third, all tests used a single device model (Samsung Galaxy A54). Lower-end devices with less RAM, weaker GPUs, and smaller screens (common in the target demographic) may exhibit longer paint times, dropped frames on charts, or layout breakage not observed here.

Fourth, network conditions were simulated via Chrome DevTools throttling. Real 4G in rural Ghana exhibits higher latency variance, periodic drops to 2G, and data-cost sensitivity that a fixed throttle does not capture. The offline-first architecture mitigates this, but the AI features degrade to fallback mode when connectivity fails — a transition not tested under real field conditions.

Fifth, the evaluation covered the current feature set only. The planting calendar covers 394 lines of static data for West African crops; it has not been validated by agronomists for accuracy in each climate zone. The AI advice has not been evaluated by domain experts for agronomic correctness. Functional testing verifies that features execute; it does not verify that the advice they produce is sound.

Sixth, the study did not measure longitudinal adoption, learning effects, or impact on farming outcomes. A farmer who uses AgriFlow for a full season may develop different mental models, encounter different errors, and derive different value than a first-time user in a forty-minute session. Longitudinal field deployment is the necessary next step.

5.7 Summary

Functional testing confirmed that all fourteen requirements and their fallback paths operate as specified. Usability testing with twelve participants yielded a SUS score of 78.5 (Good) and identified three priority improvements: a pre-permission rationale for the voice agent, a more prominent scanner entry point, and reconsideration of the command palette investment. Performance testing validated the lazy-loading and offline-first architecture against all NFR-01 targets on a mid-range device over throttled 4G. The limitations section acknowledges the small sample, narrow demographic, single device, simulated network, unvalidated agronomic content, and absence of longitudinal data. These findings direct the next iteration toward voice-agent onboarding, scanner discoverability, multi-device testing, and a field trial with agronomic validation.